# PRATHAM JAISWAL

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## SUMMARY

Systems and automation engineer. 22 merged pull requests upstream into rust-lang/rustc_codegen_cranelift, google-deepmind/open_spiel, pytorch/ao, CCExtractor and CircuitVerse, with further work open against rust-clippy, wagtail and gatsby. Currently builds ticket-automation and vulnerability-reporting systems for a distributed healthcare research organisation. Rust, TypeScript, Python, PostgreSQL, Docker.

## OPEN SOURCE

**rust-lang/rustc_codegen_cranelift** — compiler backend. PR 1625 (merged): ported an optimised disk-cleanup workflow that resolved CI runner disk exhaustion and restored abi-cafe test reliability. PR 1711 (open): fixed an internal compiler error on const asm operands pointing at anonymous allocations by materialising the allocation and threading the module into GlobalAsmContext.

**CCExtractor/ccextractor** — subtitle extraction, C. PR 2378 (merged): root-caused wrong tx3g subtitle timings on the FFmpeg MP4 path; the demuxer derived times from a shared timing pointer that interleaved tracks had already advanced, so both paths now use packet PTS plus sample duration.

**google-deepmind/open_spiel** — PR 1455 (merged): replaced legacy TensorFlow 1.x deep reinforcement learning examples with modern PyTorch equivalents.

**pytorch/ao** — PR 3709 (merged): documented FqnToConfig and mixture-of-experts parameter-quantisation workflows in the API reference.

**CircuitVerse/cv-frontend-vue** — six merged: removed jQuery from engine, setup and minimap modules; converted subcircuit, events, sequential and LSB/MSB modules to TypeScript during the v1 simulator migration.

**Also merged:** Razorpay payment flow (homxmagic), telemetry and persistent queue storage (scheduler), video quality scoring (anti-y-web), Google OAuth and JWT decoding (lost-and-found), signup fixes (profileFolio).

**Open:** gatsby-plugin-manifest assetPrefix fix (gatsbyjs/gatsby PR 39703); OSGi and .bnd dependency parsing (aboutcode-org/scancode-toolkit PRs 4765, 4766).

## EXPERIENCE

**Braid-Forbes Health Research** — Full-Stack Developer, IT Systems and Operations | Remote | Apr 2026 to Present

- Built a ticket automation layer over monday.com and Outlook: intake through received, on-hold and completion transitions with state-driven notifications, including debugging of sender connections and template state.
- Built a vulnerability reporting pipeline that consolidates endpoint findings by application, severity and remediation path from ThreatDown and Malwarebytes Nebula output.
- Administer Microsoft 365 for distributed research staff: Exchange, Teams, shared mailboxes, licensing, onboarding and offboarding.
- Support remote staff across FortiClient VPN, FortiToken MFA, RDP, Windows virtual machines and NAS; document repeatable fixes.

## PROJECTS

**Perfume Store** — production commerce system (TypeScript, Next.js). Catalogue, admin console, dispatch and tracking, India Post integration, variant pricing. Over 60 reviewed pull requests. github.com/pantha704/perfume-store

**FlowForge** — serverless automation (Next.js, TypeScript, Upstash QStash). Separates frontend events from asynchronous background jobs with durable delivery and retry.

**Turbin3, Solana** — Q4 2025 Builder and Q1 2026 Accelerate (Rust, Anchor). Built NFT marketplaces, staking protocols, vaults, escrow systems and automated market makers using PDAs, CPIs and SPL tokens.

**Homeworker** — local-first document pipeline (Python). PDF and image input to reviewable A4 notes in licensed handwriting typefaces. Public repository, v0.2.0. github.com/pantha704/home-worker

**MacFolio** — browser-based macOS simulation (React, TypeScript, Zustand, GSAP). Draggable, resizable windows with Finder, Safari, Gallery and Terminal state. mac-folio-three.vercel.app

## SKILLS

**Languages:** Rust (compiler internals), TypeScript, Python, Go, Solidity, SQL **Engineering:** Next.js, React, Node.js, Anchor, Prisma, PostgreSQL, Redis, Docker, Linux, GitHub Actions **Automation:** n8n, monday.com, Upstash QStash, REST, webhooks, OAuth, Playwright, async job design **Systems:** Microsoft 365, Exchange, ThreatDown Nebula, FortiClient, FortiToken, RDP, Windows VMs

## EDUCATION

**Sister Nivedita University** — B.Tech, 2023 to 2027, Kolkata. **National English School** — ISC Science (PCM), 85 percent, 2023